



MMSG DST STORAGE FACILITY

MCB CAMP LEJEUNE, NORTH CAROLINA

Geotechnical Engineering Report

PREPARED BY:



151 Poole Road, Suite 100

Leland, NC 28451

(910) 383-1044

NC License C-1621

PREPARED FOR:

PUBLIC WORKS DIVISION

MCB CAMP LEJEUNE, NC

PROJECT NO. G2023-186

DECEMBER 18, 2023





December 18, 2023

Commanding General
Attn: Public Works Division
Joe Cirillo, Project Manager
MCB, PSC Box 20004
Camp Lejeune, NC 28542-0004

RE: Geotechnical Engineering Report
23-0033 Construct MRSG DST Storage Facility MARSOC
MCB Camp Lejeune, North Carolina
Project No: G2023-186

Dear Mr. Cirillo:

Cape Fear Engineering, Inc. (CFE) has completed the geotechnical engineering services for the above referenced project. This report presents the results of the subsurface exploration and provides our geotechnical engineering recommendations.

We appreciate the opportunity to provide our services to you on this project. Should you have any questions or if we can be of further assistance, please contact us.

Respectfully Submitted,
Cape Fear Engineering

Michael C. Raup, P.E.
Project Engineer
NC Reg. # 045271



TABLE OF CONTENTS

1.0 INTRODUCTION	1
1.1 Project Site Location and Description	1
1.2 Scope of Services	2
1.3 Project Authorization	2
2.0 FIELD EXPLORATION AND LABORATORY TESTING.....	3
2.1 Field Exploration	3
2.2 Laboratory Testing	4
3.0 SUBSURFACE CONDITIONS.....	5
3.1 Site Geology	5
3.2 Subsurface Soil Conditions	5
3.3 Groundwater.....	6
4.0 DESIGN AND CONSTRUCTION RECOMMENDATIONS.....	7
4.1 Earthwork - Site Clearing and Grading.....	7
4.2 Earthwork - Subgrade Preparation and Evaluation.....	8
4.3 Earthwork – Suitable Structural Fill, Placement, and Compaction Requirements	9
4.4 Foundations.....	10
4.5 Building Floor Slabs	12
4.7 Seismic Evaluation.....	12
4.8 Construction Considerations	12
5.0 REPORT LIMITATIONS.....	15
 APPENDIX I	BORING LOCATION EXHIBIT
APPENDIX II	CLASSIFICATION SYSTEM FOR SOIL EXPLORATION
APPENDIX III	BORING LOGS
APPENDIX IV	SOIL PROFILE

1.0 INTRODUCTION

1.1 Project Site Location and Description

Cape Fear Engineering has completed our geotechnical engineering services for the proposed new storage building to be constructed at the Construct MRSG DST Storage Facility within the MARSOC section of the Marine Corps Base (MCB) Camp Lejeune military installation in North Carolina. It is our understanding that the 22-0033 Construct MRSG DST Storage Facility project will include the construction of a 1-story pre-engineered metal building with a footprint of 100-feet by 60-feet. The purpose of this geotechnical engineering evaluation was to determine pertinent subsurface information for earthwork and foundation design considerations.

The proposed pre-engineered metal building will consist of steel frame, a standing seam metal roof, and a concrete monolithic slab-on-grade with turn down edges. The maximum wall and column loads associated with the new structure are not expected to exceed 1.5 kips per linear foot and 65 kips, respectively.

Based on the previously discussed project information, documents provided by the client, and observations during our site reconnaissance, site elevations within the project area are anticipated to be at or near current elevations. Therefore, we anticipate that less than 1-foot of fill is expected to establish final site grades.

The project site is located in a wooded tract with minimal grade grades across the project location. The surrounding area includes existing MARSOC military facilities, parking areas, existing roadways, open and wooded tracts of land.

A site vicinity map showing the project area is provided below.



Project Site General Vicinity

1.2 Scope of Services

The purpose of this investigation was to obtain information on the general subsurface conditions within the project area. The subsurface conditions were evaluated to provide our engineering assessments. For this project, the following items were evaluated to provide geotechnical engineering information and recommendations:

- ❖ General assessment of the soils revealed by the borings performed at the project site, as well as provide the depth of groundwater at the boring locations at the time of drilling.
- ❖ General location and description of any potentially deleterious materials encountered in the borings that may interfere with the construction process or structural performance of the proposed structure. Potential deleterious materials include existing fills, expansive soils, substantial organics, or other unsuitable materials.
- ❖ Interpretation of the soil test borings to provide earthwork design and construction recommendations including stripping, grading, engineering requirements for structural fill, placement, and compaction.
- ❖ Interpretation of the soil test borings to provide recommendations for a shallow foundation system and other foundation construction recommendations for support of the proposed structure. Design and construction requirements including allowable bearing pressures, foundation sizes, foundation embedment, expected total and differential settlements, foundation excavation preparation, and required foundation observation and testing.
- ❖ Seismic site class determination in accordance with the International Building Code (IBC).

The scope of services did not include an environmental assessment for determining the presence or absence of wetlands, hazardous or toxic material in the soil, bedrock, surface water, groundwater, or air on, below, or in the vicinity of the project site.

1.3 Project Authorization

The Geotechnical Engineering Services were conducted in general accordance with the Cape Fear Engineering proposed services. Authorization to proceed with our services was received from Mr. Joe Cirillo of the MCB Camp Lejeune Public Works Division.

2.0 FIELD EXPLORATION AND LABORATORY TESTING

2.1 Field Exploration

The general subsurface soil types were explored by completing Standard Penetration Tests (SPT) in the areas noted in Table I below. The information obtained from our field exploration program was used to assist in developing the design and construction recommendations.

Table I – Boring Schedule

Boring Number	Boring Depth ⁽¹⁾ (feet)	Boring Location Description
B-1	30.0	Building boring located in the approximate southern corner of the proposed building footprint.
B-2	40.0	Building boring located in the approximate northwestern corner of the proposed building footprint.
B-3	50.0	Drive lane boring located near the proposed building footprint.

⁽¹⁾ Boring depth measured from the existing ground elevation at time of drilling.

Standard Penetration Tests were performed using mud rotary techniques with a drill rig in general accordance with ASTM D1586 and ASTM D5783. The tests were performed continuously from the existing ground surface to a depth of 12 feet and at 5-foot intervals, thereafter, starting at a depth of 13 feet below the existing site grade. The soil samples were obtained with a standard 30" long split-spoon sampler. The sampler was driven with blows of a 140-pound automatic hammer falling 30 inches. The number of blows required to drive the sampler through each 6-inch increment of penetration was recorded and is shown on the boring log. The sum of the second and third penetration increments is termed the SPT N-value (uncorrected for automatic hammer and overburden pressure). A representative portion of each disturbed split-spoon sample was collected with each SPT, placed in a glass jar, sealed, labeled, and returned to our laboratory for review.

The boring locations were initially established by a representative of Cape Fear Engineering. The approximate boring locations are shown on the attached "Boring Location Exhibit" (Appendix I), which was reproduced from a Site Plan prepared by Cape Fear Engineering.

2.2 Laboratory Testing

Soil testing provided by Cape Fear Engineering was performed in accordance with American Society for Testing and Materials (ASTM) standards. All soil testing was performed in our Leland, North Carolina laboratory.

Representative portions of all soil samples collected during drilling operations were labeled, sealed in glass jars, and transferred to our laboratory in accordance with ASTM D4220 for classification and analysis. Soil descriptions on the boring logs are provided in general accordance with ASTM D2488 using the Unified Soil Classification System (USCS.) Soil samples that were selected for testing were classified in general accordance with ASTM D2487. Some variation can be expected between samples classified using the visual-manual procedure (ASTM D2488) and the USCS (ASTM D2487.) A summary of the soil classification system is provided in Appendix II.

As indicated above, representative soil samples were selected and subjected to natural moisture, No. 200 sieve wash, and Atterberg Limits testing to verify the visual classification. These test results are tabulated below in Table II – Summary of Laboratory Test Results below. These results are also presented on the Boring Logs provided in Appendix III. A generalized subsurface Soil Profile is also provided in Appendix IV.

Table II – Summary of Laboratory Test Results

Boring Number	Sample Type	Depth (feet)	Natural Moisture (%)	Passing No. 200 Sieve (%)	Atterberg Limits (LL/PL/PI)	USCS Classification
B-1	Split Spoon	2-4	12.5	16.2	Non-Plastic	Silty fine to coarse SAND (SM) with trace Gravel
B-1	Split Spoon	18.5-20.0	37.3	35.7	Non-Plastic	Silty fine to medium SAND (SM) with trace Gravel
B-2	Split Spoon	2-4	13.2	27.9	Non-Plastic	Silty fine to coarse SAND (SM) with Gravel
B-2	Split Spoon	13.5-15.0	39.4	32.7	Non-Plastic	Silty fine to medium SAND (SM)
B-3	Split Spoon	0-2	15.0	9.1	Non-Plastic	Poorly Graded fine to medium SAND (SP-SM) with Silt and trace Gravel
P-1	Split Spoon	6-8	25.4	21.7	Non-Plastic	Silty fine to medium SAND (SM)
P-2	Split Spoon	28.5-30.0	43.4	46.6	Non-Plastic	Silty fine to medium SAND (SM)

3.0 SUBSURFACE CONDITIONS

3.1 Site Geology

The project site lies within a major physiographic province of North Carolina called the Atlantic Coastal Plain. Numerous transgressions and regressions of the Atlantic Ocean have deposited marine, lagoonal, and fluvial (stream lain) sediments. The regional geology is very complex, and generally consists of interbedded layers of varying mixtures of sands, silts, and clays. Based on our review of existing geologic and soil boring data, the geologic stratigraphy encountered in our subsurface exploration generally consisted of marine deposited sands and clays.

3.2 Subsurface Soil Conditions

A summary of the subsurface soil conditions encountered at the boring locations is presented in Table III.

Table III – Subsurface Soil Conditions

Average Depth (ft.)	Stratum	Description	Range of SPT ⁽¹⁾ Values
0.0 to 0.2-0.5	Surficial	2 to 6 inches of Topsoil	-
0.2-0.6 to 50.0	I	Dark brown, grayish brown, brown, gray, brown and gray, light grayish brown, light brown, dark green and light gray, light grayish brown, SAND (SP-SM, SM) with varying amounts of silt and clay.	<u>Granular</u> WOH ⁽²⁾ – 100
Note(s): (1) SPT = Standard Penetration Test, N-Values in blows per foot (uncorrected) (2) WOH = Weight of Hammer			

The subsurface descriptions are of a generalized nature and were provided to highlight the major soil strata encountered. The records of the subsurface exploration are included in Appendix III (Boring Log sheets) and in Appendix IV (Soil Profile), which should be reviewed for specific information as to the individual borings. The stratifications shown on the record of the subsurface exploration represent the conditions only at the actual boring location. Variations may occur and should be expected across the site. The stratifications represent the approximate boundary between subsurface materials and the transition may be gradual.

3.3 Groundwater

The initial groundwater level was recorded at the boring locations and as observed through the wetness of the recovered soil samples during the drilling operations. The initial groundwater table was measured to occur at a depth of approximately 5.0 – 8.0 feet below the existing site grade at boring locations. Any variations in groundwater readings are anticipated to be the result of the effects of rain events, associated man-made disturbances, and drainage features. The borehole was backfilled with grout upon completion for safety considerations. Seasonal groundwater fluctuations of $2\pm$ feet or more are common in the project's area; however, greater fluctuations have been documented. The contractor should determine the actual groundwater levels at the time of construction to determine groundwater impacts that may affect the construction procedures.

Perched groundwater conditions may be encountered at the project site. The perched condition is anticipated to occur during periods of heavy precipitation and/or during the wet season in the areas where shallow subsurface clayey soils were encountered. These soils will act as a restrictive layer allowing excessive moisture to accumulate within the overlying granular soils. Perched groundwater, if encountered, can adversely affect foundation excavation and other construction activities. The contractor should determine the actual groundwater levels and potential perched groundwater conditions at the time of construction to determine groundwater and potential perched groundwater impacts that may affect the construction procedures.

4.0 DESIGN AND CONSTRUCTION RECOMMENDATIONS

Our design and construction recommendations are based on the previously discussed project information, interpretation of the soil test borings, review of the documents provided by the client, and observations during our site reconnaissance. If the proposed construction should vary from what was described, we request the opportunity to review our recommendations and make any necessary changes.

4.1 Earthwork - Site Clearing and Grading

The proposed new construction area (new building pad) should be cleared by removing any topsoil or associated root mat and any other unsuitable material, if encountered. Based on the SPT boring, 2 to 6 inches of topsoil was encountered at the time of drilling. However, the project site near previously constructed building and is expected to contain varying amounts of unsuitable material associated previous construction activities. It is recommended that the clearing operations extend laterally at least 5 feet beyond the perimeter of the proposed construction area.

Once the site clearing is completed, the exposed subgrade will generally be comprised of very loose to medium dense SAND (SP-SM, SM). Due to the presence of very loose to medium dense SAND, excess surface moisture from precipitation ponding on the site, along with construction equipment traffic, may make this site susceptible to pumping and general deterioration of the bearing capabilities of the surface soils. Therefore, undercutting to remove wet soils may be required. The extent of the undercut will be determined in the field during construction, based on the outcome of the field-testing procedures (subgrade proofroll and test pits). The project's budget should include an allowance for subgrade improvements (building undercut and backfill with structural fill).

To reduce the potential for subgrade improvements, it is recommended that the grading operations be performed during the drier months of the year (generally April through November.) This should help minimize these potential problems. If grading is attempted during the winter months, the site should be graded to enhance surface water runoff. However, stabilization of wet soils should be anticipated. Methods to address wet soils may include undercutting and backfilling with structural fill. However, during the drier months of the year, wet soils could be dried by disking or implementing other drying procedures such as stockpiling or spreading the soils in thin lifts, to achieve moisture contents necessary to reach adequate degrees of compaction. As previously stated, the project's budget should include an allowance for subgrade improvements as described above.

Any undercut and backfill should be performed under the observation of the geotechnical engineer or their qualified representative who will evaluate the composition of the recovered soils. Recommendations concerning the subgrade improvements (as necessary) will be provided in the field following the testing procedures.

As previously indicated, perched groundwater conditions will likely be encountered at this site. The perched condition is anticipated to occur during periods of heavy precipitation and/or during the wet season in the areas where shallow subsurface and/or surface clayey soils were encountered. These soils will act as a restrictive layer allowing excessive moisture to accumulate at the surface or within the overlying granular soils. Perched groundwater, if encountered, can adversely affect foundation excavation, roadway, utility installation, and other construction activities. The contractor should determine the actual groundwater levels and potential perched groundwater conditions at the time of construction to determine groundwater and potential perched groundwater impacts that may affect the construction procedures.

4.2 Earthwork - Subgrade Preparation and Evaluation

Following the clearing operation, the exposed subgrade soils should be densified with a large static drum roller. After the subgrade soils have been densified, they should be evaluated by a qualified representative of Cape Fear Engineering or the Contractor's Geotechnical Engineer for stability. The subgrade soils should be proofrolled to check for pockets of soft material hidden beneath a crust of better soil. Several passes should be made by a large roller compactor (with at least 2-rubber tires) or a loaded dump truck over the construction areas, with the successive passes aligned perpendicularly. The number of passes will be determined in the field by the geotechnical engineer or their qualified representative. Any pumping and unstable areas observed during proofrolling (beyond the initial clearing cut) should be undercut and/or stabilized at the direction of the geotechnical engineer or their qualified representative.

In addition to the proofroll, a series of test pit excavations should be performed to determine the extent of any potential uncontrolled earthfill or other unsuitable materials which could become evident during the site stripping operation within the proposed building pad and other structural construction areas. The test pit excavations should be performed under the observation of the geotechnical engineer or their qualified representative to determine the thickness and composition of any Possible Uncontrolled Earthfill materials or other unsuitable materials and the suitability of the materials to remain in place or the necessity for these materials to be removed from the building and other structural construction areas.

The prepared subgrade should be sloped to prevent the accumulation and/or ponding of surface water. If the exposed subgrade becomes wet or frozen, the geotechnical engineer should be consulted.

4.3 Earthwork – Suitable Structural Fill, Placement, and Compaction Requirements

Following the approval of the natural subgrade soils by a geotechnical engineer or their qualified representative, the placement of the fill required to establish the design grades may begin. Any material to be used for structural fill should be evaluated and tested by a qualified inspector and laboratory prior to placement to determine if they are suitable for the intended use. Suitable structural fill material should consist of sand or gravel containing less than 20% by weight of fines (SP, SP-SM, SM, SW, SW-SM, GP, GP-GM, GW, GW-GM), have a liquid limit less than 20 and plastic limit less than 6, and should be free of rubble, organics, clay, debris, and other unsuitable material.

All structural fills should be compacted to a dry density of at least 95% of the Modified Proctor maximum dry density (ASTM D1557). In general, the compaction should be accomplished by placing the fill in maximum 10-inch loose lifts and mechanically compacting each lift to at least the specified minimum dry density. A qualified inspector should perform field density tests on each lift as necessary to assure that adequate compaction is achieved.

Backfill material in utility trenches within the construction areas should consist of structural fill and be compacted to at least 95% of ASTM D1557. This fill should be placed in 4 to 6-inch loose lifts when hand compaction equipment is used.

Care should be used when operating the compactors near existing structures to avoid transmission of the vibrations that could cause settlement damage or disturb occupants. In this regard, it is recommended that large vibratory rollers remain at least 25 feet away from existing structures. Areas within 25 feet of existing structures should be compacted with small, hand-operated compaction equipment.

Based on the completed laboratory testing, the shallow subsurface SANDS (SP-SM) appear to meet the criteria recommended in this report for reuse as structural fill. The SAND (SM) soils encountered at the boring locations do not appear to meet the criteria recommended in this report for reuse as structural fill; however, they may be used as fill in non-structural green areas.

At a minimum, further classification testing including natural moisture content, No. 200 sieve wash analysis, and Proctor testing should be performed at the time of construction to evaluate the suitability of the excavated soils for reuse as structural fill.

4.4 Foundations

Based on the laboratory classification results, the shallow subsurface soils (upper 4-feet) encountered at the boring locations are not considered to be expansive in accordance with 1803.5.3 of the 2018 IBC. The shrink-swell potential of these subsurface soil conditions does not need to be considered and so are not included as part of our foundation recommendations.

Shallow Foundations

Provided that the construction procedures provided in Section 4.0 are properly performed, the proposed structure can be supported by isolated or continuous spread footings or a monolithic slab with turn down edges, bearing upon firm natural soil or well-compacted structural fill material. These footings can be designed using a net allowable soil pressure of 2,000 pounds per square foot (psf). In using net pressures, the weight of the footings and backfill over the footings, need not be considered. Hence, only loads applied at or above the foundation need to be used for dimensioning the footings.

To develop the recommended bearing capacity of 2,000 pounds per square foot (psf), the base of the footings should have a minimum embedment of 18 inches beneath finished grades and should have a minimum width of 18 inches. In addition, isolated square column footings (if deemed necessary) are recommended to be a minimum of 3 feet by 3 feet in area for bearing capacity consideration. The recommended 18-inch footing embedment is also considered sufficient to provide adequate cover against frost penetration to the bearing soils.

The recommended allowable bearing pressures for the shallow foundation system proposed for the new structure may potentially impact any existing underground utilities located below the building footprint. The additional loads produced by the new structure may detrimentally affect any potential underground utility facilities. Therefore, we recommend that a thorough utility locate be conducted specifically within the building footprint to determine if any underground utilities exist in this area. The building load impacts to the underground utilities, if encountered, should also be evaluated by the structural and/or civil engineer for the project to determine if the existing utility facilities can remain in place or need to be removed and relocated.

Settlements

It is estimated that, with properly performed site preparation and construction procedures as described within this report, the maximum resulting post-construction total settlement of the proposed foundations should be up to 1 inch. The maximum differential settlement magnitude is expected to be less than ½ inch between adjacent footings (column footings of varying loading conditions). The settlements were estimated based on the results of the field penetration tests. Careful field control will contribute substantially towards minimizing the settlements.

Foundation Excavations

In preparation for shallow foundation support, the footing excavations should extend into firm natural soil or well-compacted structural fill. The foundation bearing capacities should be verified in the field during construction by performing a foundation inspection for the structure. At that time, the geotechnical engineer or a qualified representative should also explore the extent of any excessively loose, soft, or any otherwise unsuitable material within the exposed excavations. Also, at the time of footing observations, the geotechnical engineer or qualified representative should advance hand auger borings and use a hand penetration device in the base of the foundation excavations to verify that the recovered soils are consistent with those documented in this report. The necessary depth of penetration will be established during the footing subgrade observations.

If pockets of unsuitable soils requiring undercut are encountered in the footing excavations, the proposed footing elevation should be re-established by means of backfilling with No. 57 Stone, "flowable fill", or lean concrete prior to concrete placement. This construction procedure will provide for a net allowable bearing capacity of 2,000 psf.

Immediately prior to reinforcing steel placement, it is suggested that the bearing surfaces of all footing and floor slab areas be compacted using hand operated mechanical tampers to a dry density of at least 95% of the Modified Proctor maximum dry density (ASTM D1557) as tested to a depth of 8 to 12 inches for bearing capacity considerations. In this manner, any localized areas which have been loosened by excavation operations should be adequately re-compacted. The compaction testing in the base of the footings may be waived by the geotechnical engineer, where firm bearing soils are observed during the footing inspections.

Soils exposed in the bases of all satisfactory foundation excavations should be protected against any detrimental change in condition such as from physical disturbance, rain, or frost. Surface run-off water should be drained away from the excavations and not be allowed to pond. If possible, all footing concrete should be placed the same day the excavation is made. If this is not possible, the footing excavations should be adequately protected.

As previously indicated, perched groundwater conditions will likely be encountered at this site. The perched condition is anticipated to occur during periods of heavy precipitation and/or during the wet season in the areas where shallow subsurface and/or surface clayey soils were encountered. These soils will act as a restrictive layer allowing excessive moisture to accumulate at the surface or within the overlying granular soils. Perched groundwater, if encountered, can adversely affect foundation excavation, roadway, utility installation, and other construction activities. The contractor should determine the actual groundwater levels and potential perched groundwater conditions at the time of construction to determine groundwater and potential perched groundwater impacts that may affect the construction procedures.

4.5 Building Floor Slabs

The building floor slab may be constructed as slab-on-grade members provided the previously recommended earthwork activities and evaluations are carried out properly. It is recommended that the ground floor slab be directly supported by at least a 4-inch layer of relatively clean, compacted, poorly graded sand (SP) or gravel (GP) with less than 5% passing the No. 200 Sieve (0.074 mm). The purpose of the 4-inch layer is to act as a capillary barrier and equalize moisture conditions beneath the slab. The slabs can be designed with the use of a subgrade modulus on the order of about 125 psi/in bearing on the existing natural subgrade soils or bearing on structural fill compacted to 95 percent of the Modified Proctor maximum dry density (ASTM D1557).

It is also recommended that the floor slab bearing soils be covered by a vapor barrier or retarder to minimize the potential for floor dampness, which can affect the performance of glued tile and carpet. When floor finishes, site conditions, or other considerations require greater vapor resistance protection, consideration should be given to using a vapor barrier. Selection of a vapor retarder or barrier should be made by the architect based on project requirements.

4.7 Seismic Evaluation

Based on the data obtained from the 35 to 50-foot deep SPT boring and our experience with 100-foot-deep CPT soundings and SPT borings performed within the vicinity of the project site, this site should be classified as a Site Class 'D' in accordance with the IBC.

4.8 Construction Considerations

Based on the results of this exploration, varying soil conditions and compositions are expected to be encountered throughout the project limits. Open-cut excavations will likely extend through natural soils that are relatively "clean" (i.e., soil that is free of deleterious debris that may hinder excavation or installation). Debris typically considered unsuitable consist of wood, glass, organics, plastics, coal, brick, or any other material larger than 2 inches in diameter. Based on these characteristics, it is anticipated that some of the shallow subsurface materials encountered within the project alignment may be reusable as backfill. Soils containing appreciable amounts of deleterious debris should be discarded; however, an effort should be made during excavation to segregate potentially suitable in-situ fill soils for reuse.

The shallow subsurface within the project limits is comprised of granular soils; however, the contractor should anticipate that some of these soils will have relatively little cohesion and have a high potential for caving. Additionally, water seepage at varying elevations should be expected within the side walls of the open cut areas, increasing the potential for caving.

Temporary Slopes

Due to the limited space for construction, temporary slopes may not be a feasible option. The contractor should be aware that temporary slope height, slope inclination, or excavation depths should in no case exceed those specified in local, state and/or federal safety regulations. Where temporary slopes are not feasible, shoring by means of sheeting and/or trench boxes may be appropriate. Where the stability of adjoining structures, pavements, or other improvements is endangered by excavation operations, support systems such as shoring, bracing, or underpinning may be required to provide structural stability. Shoring, bracing, or underpinning required for this project (if required) shall be designed by a professional engineer.

Shoring

Shoring design and installation should be the responsibility of the contractor. Shoring systems required for this project should be designed by a professional engineer. Shoring systems should be designed to provide positive restraint of trench walls to protect against lateral deformation that may result in ground cracks, settlement, and/or other ground movements that may affect adjacent underground utilities, pavements, structures, and surface improvements. The contractor should be made aware of this potential condition so that preventative or repair measures can be implemented.

Depending on the shoring system used, the removal process may create voids along the walls of the excavations. If these voids are left in place and are significant, backfill and/or the retained soil may shift laterally, resulting in settlement of overlying structures/pavements. As such, care should be taken to remove the shoring systems and backfilling the trenches so that these voids are not created.

In all cases, the contractor should select an excavation and/or shoring scheme that will protect adjacent and overlying improvements, including below grade utilities.

Dewatering

It is expected that dewatering will be required for excavations that extend near or below the existing groundwater table. Dewatering above the groundwater-level could probably be accomplished by pumping from sumps. Dewatering at depths below the groundwater-level will require well pointing and possibly shoring. Since temporary dewatering will impact construction and be dependent on construction methods and scheduling, we recommend the contractor be solely responsible for the design, installation, maintenance, and performance of all temporary dewatering systems. Where shoring is employed, the dewatering system should be compatible with the type of shoring to be used. We recommend the Contractor verify groundwater conditions and evaluate dewatering requirements prior to construction.

Lowering the groundwater table during dewatering activities will result in an increase in effective stresses and may induce settlements of the soils underlying adjacent structures/pavements. Additionally, hydraulic compaction of the predominately granular soils (e.g., SP-SM, SM soils) should be anticipated because of lowering the groundwater table. We recommend that the dewatering be performed so that the groundwater-level is lowered by no more than approximately 5 feet below the proposed excavation depth. It may be advantageous to install settlement monuments in areas where dewatering by means of well pointing is required.

Site Utility Installation

The base of the utility trenches should be observed by a qualified inspector prior to the pipe placement to verify the suitability of the bearing soils. Based on the results of our field exploration program, it is expected that the utilities located below the groundwater level will bear in the wet granular soils. In these instances, the bearing soils may require some stabilization to provide suitable bedding. This stabilization is commonly accomplished by adding 12 inches or more of bedding stone (Type NCDOT #57). The resulting excavations located within structural areas should be backfilled with structural fill, provided the construction procedures provided in Section 4.3 are properly performed. Imported fill should be included in the construction budget to backfill the utility excavations within the structural areas of the project site.

Excavations

Federal regulation requires all excavations, whether they be utility trenches, basement excavation, or footing excavations, be constructed in accordance with (OSHA) guidelines.

The contractor is solely responsible for designing and constructing stable, temporary excavations and should shore, slope, or bench the sides of the excavations as required to maintain stability of both the excavation sides and bottom. The contractor's responsible person should evaluate the soil exposed in the excavations as part of the contractor's safety procedures. In no case should slope height, slope inclination, or excavation depth, including utility trench excavation depth, exceed those specified in local, state, and federal safety regulations.

5.0 REPORT LIMITATIONS

The recommendations submitted are based on the available soil information obtained by Cape Fear Engineering and the information supplied by the client for the proposed project. If there are any revisions to the plans for this project or if deviations from the subsurface conditions noted in this report are encountered during construction, Cape Fear Engineering should be notified immediately to determine if changes in the recommendations are warranted. If Cape Fear Engineering is not retained to perform these functions, Cape Fear Engineering cannot be responsible for the impact of those conditions on the geotechnical recommendations for the project.

The geotechnical engineer of record warrants that the findings, recommendations, specifications, or professional advice contained herein have been made in accordance with generally accepted professional geotechnical engineering practices in the local area. No other warranties are implied or expressed.

After the plans and specifications are more complete, the geotechnical engineer of record should be provided the opportunity to review the final design plans and specifications to make sure our engineering recommendations have been properly incorporated into the design documents, in order that the earthwork and foundation recommendations may be properly interpreted and implemented. At that time, it may be necessary to submit supplementary recommendations.

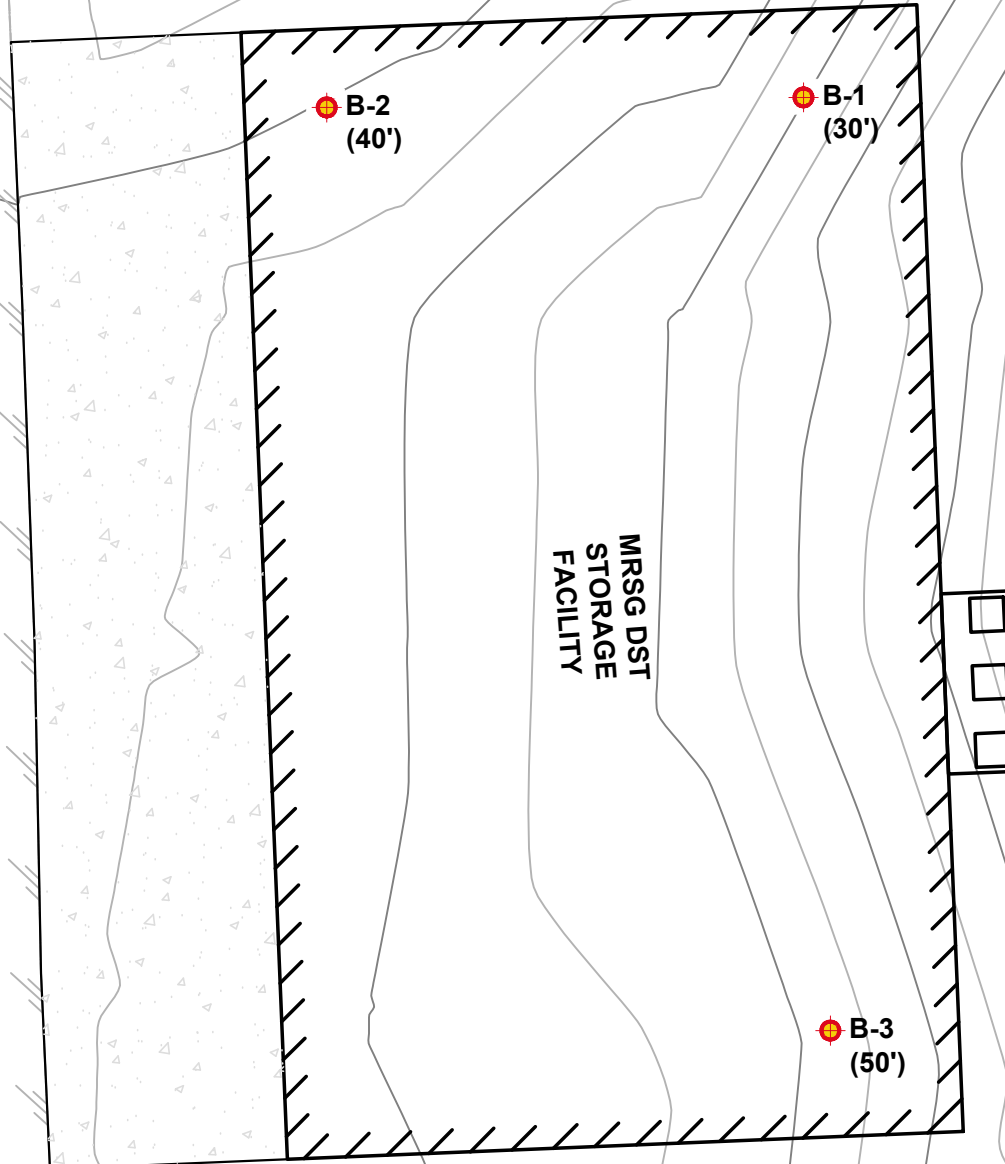
This report has been prepared for the exclusive use of the client and their designated agents for the specific application to the 23-0033 Construct MRSG DST Storage Facility project located in the MARSOC section of MCB Camp Lejeune military installation in North Carolina.

APPENDICES

APPENDIX I	BORING LOCATION EXHIBIT
APPENDIX II	CLASSIFICATION SYSTEM FOR SOIL EXPLORATION
APPENDIX III	BORING LOGS
APPENDIX IV	SOIL PROFILE
APPENDIX V	LABORATORY TEST RESULTS

APPENDIX I

BORING LOCATION EXHIBIT



VICINITY MAP
NOT TO SCALE

LEGEND

 "B" BORING LOCATIONS
(30-50 FEET)

NOTES

1. THIS BORING LOCATION EXHIBIT WAS REPRODUCED FROM A PLAN PROVIDED BY THE CLIENT.
2. THIS BORING EXHIBIT IS NOT TO SCALE.
3. ALL BORING LOCATIONS ARE APPROXIMATE.

PREPARED FOR:
PUBLIC WORKS DIVISION



CAPE FEAR ENGINEERING
151 Poole Rd. Suite 100 | Leland, NC, 28451
TEL (910) 383-1044 | FAX (910) 383-1045
www.capefearengineering.com | N.C. LICENSE # C-1621

DATE: 12-18-2023

SCALE: NOT TO SCALE

DRAWN: JGC

APPROVED: MCR

23-0033 CONSTRUCT MRSG DST
STORAGE FACILITY MARSOC
MCB CAMP LEJEUNE, NC

BORING LOCATION
EXHIBIT

JOB NUMBER
G2023-186

SHEET NUMBER
EX-01

APPENDIX II

CLASSIFICATION SYSTEM FOR SOIL EXPLORATION



CLASSIFICATION SYSTEM FOR SOIL EXPLORATION

Standard Penetration Test (SPT), N-value

Standard Penetration Tests (SPT) were performed in the field in general accordance with ASTM D 1586. The soil samples were obtained with a standard 1.4" I.D., 2" O.D., 30" long split-spoon sampler. The sampler was driven with blows of a 140 lb. hammer falling 30 inches. The number of blows required to drive the sampler each 6-inch increment (4 increments for each soil sample) of penetration was recorded and is shown on the boring logs. The sum of the second and third penetration increments is termed the SPT N-value.

NON COHESIVE SOILS

(SILT, SAND, GRAVEL and Combinations)

Relative Density

Very Loose	4 blows/ft. or less
Loose	5 to 10 blows/ft.
Medium Dense	11 to 30 blows/ft.
Dense	31 to 50 blows/ft.
Very Dense	51 blows/ft. or more

Particle Size Identification

Boulders	8 inch diameter or more
Cobbles	3 to 8 inch diameter
Gravel	Coarse 1 to 3 inch diameter
	Medium $\frac{1}{2}$ to 1 inch diameter
	Fine $\frac{1}{4}$ to $\frac{1}{2}$ inch diameter
Sand	Coarse 2.00 mm to $\frac{1}{4}$ inch (diameter of pencil lead)
	Medium 0.42 to 2.00 mm (diameter of broom straw)
	Fine 0.074 to 0.42 mm (diameter of human hair)
Silt	0.002 to 0.074 mm (cannot see particles)

CLASSIFICATION SYMBOLS (ASTM D 2487 and D 2488)

Coarse Grained Soils

More than 50% retained on No. 200 sieve

GW - Well-graded Gravel
GP - Poorly graded Gravel
GW-GM - Well-graded Gravel w/Silt
GW-GC - Well-graded Gravel w/Clay
GP-GM - Poorly graded Gravel w/Silt
GP-GC - Poorly graded Gravel w/Clay
GM - Silty Gravel
GC - Clayey Gravel
GC-GM - Silty, Clayey Gravel
SW - Well-graded Sand
SP - Poorly graded Sand
SW-SM - Well-graded Sand w/Silt
SW-SC - Well-graded Sand w/Clay
SP-SM - Poorly graded Sand w/Silt
SP-SC - Poorly graded Sand w/Clay
SM - Silty Sand
SC - Clayey Sand
SC-SM - Silty, Clayey Sand

Fine-Grained Soils

50% or more passes the No. 200 sieve

CL - Lean Clay
CL-ML - Silty Clay
ML - Silt
OL - Organic Clay/Silt
 Liquid Limit 50% or greater
CH - Fat Clay
MH - Elastic Silt
OH - Organic Clay/Silt

Highly Organic Soils

PT - Peat

COHESIVE SOILS

(CLAY, SILT and Combinations)

Consistency

Very Soft	2 blows/ft. or less
Soft	3 to 4 blows/ft.
Medium Stiff	5 to 8 blows/ft.
Stiff	9 to 15 blows/ft.
Very Stiff	16 to 30 blows/ft.
Hard	31 blows/ft. or more

Relative Proportions

<u>Descriptive Term</u>	<u>Percent</u>
Trace	0-5
Few	5-10
Little	15-25
Some	30-45
Mostly	50-100

Strata Changes

In the column "Description" on the boring log, the horizontal lines represent approximate strata changes.

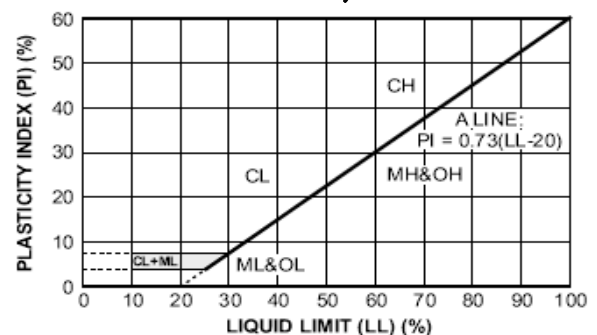
Groundwater Readings

Groundwater conditions will vary with environmental variations and seasonal conditions, such as the frequency and magnitude of rainfall patterns, as well as tidal influences and man-made influences, such as existing swales, drainage ponds, underdrains and areas of covered soil (paved parking lots, side walks, etc.).

Depending on percentage of fines (fraction smaller than No. 200 sieve size), coarse-grained soils are classified as follows:

Less than 5 percent	GW, GP, SW, SP
More than 12 percent	GM, GC, SM, SC
5 to 12 percent	Borderline cases requiring dual symbols

Plasticity Chart



APPENDIX III

BORING LOGS



BORING LOG No. B-1

PROJECT: MRSB DST Storage Project

CLIENT: CFE IDIQ

PROJECT LOCATION: MCB Camp Lejeune, NC

LOCATION: See Attached Boring Location Exhibit

DRILLER: Catlin Engineers and Scientists

DRILLING METHOD: Mud Rotary

DEPTH TO - WATER> INITIAL: 4

AFTER 24 HOURS: 6.5

PROJECT NO.: G2023-186

NORTHING:

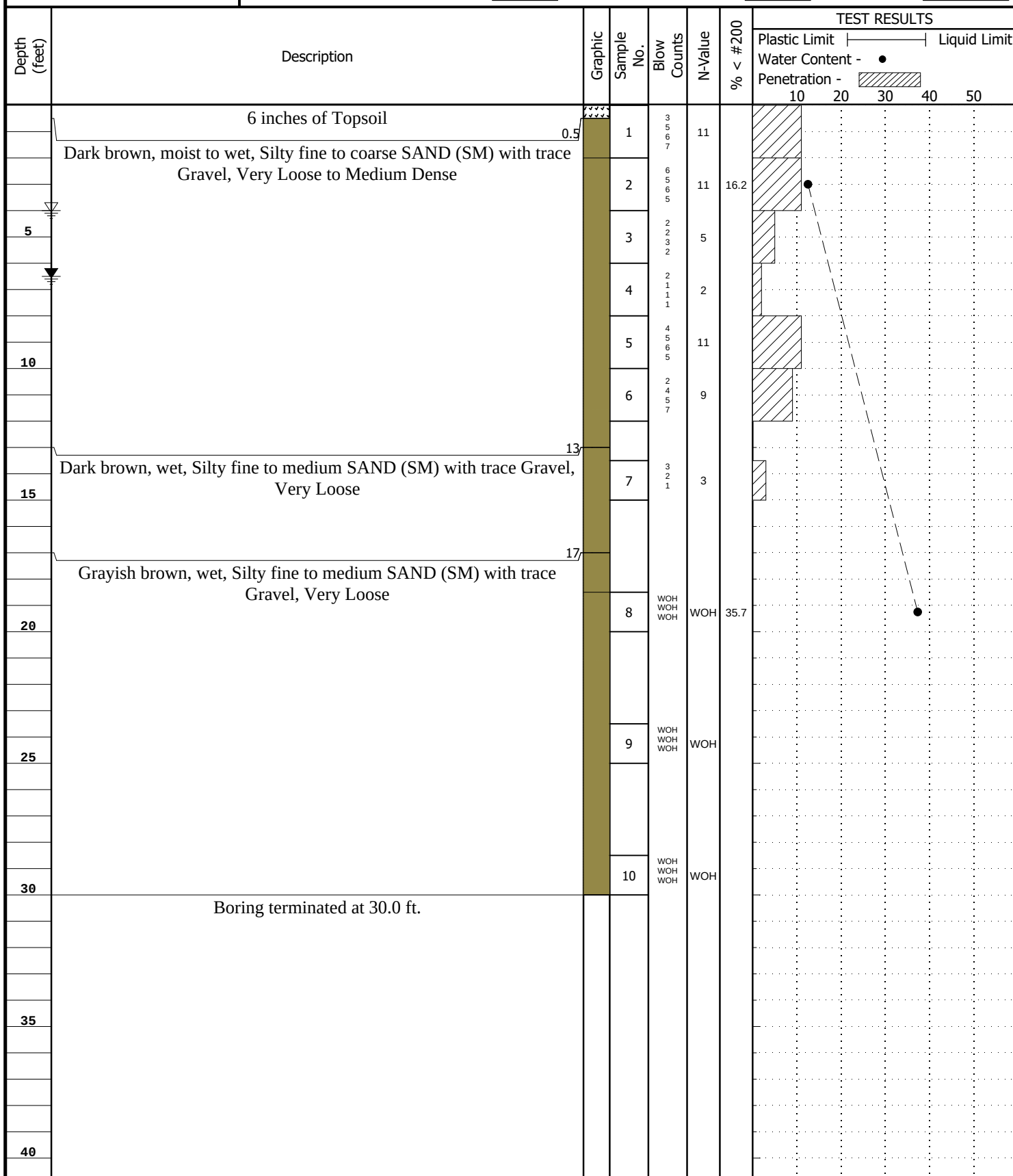
EASTING:

ELEVATION:

LOGGED BY: MCR

DATE: 11-20-23

CAVING> C



This information pertains only to this boring and should not be interpreted as being indicative of the site.



BORING LOG No. B-2

PROJECT: MRSO DST Storage Project

CLIENT: CFE IDIQ

PROJECT LOCATION: MCB Camp Lejeune, NC

LOCATION: See Attached Boring Location Exhibit

DRILLER: Catlin Engineers and Scientists

DRILLING METHOD: Mud Rotary

DEPTH TO - WATER> INITIAL: 5

AFTER 24 HOURS: 8

PROJECT NO.: G2023-186

NORTHING:

EASTING:

ELEVATION:

LOGGED BY: MCR

DATE: 11-20-23

CAVING> C

Depth (feet)	Description	Graphic Sample No.	Blow Counts	N-Value	% < #200	TEST RESULTS	
						Plastic Limit	Liquid Limit
	6 inches of Topsoil	1	5 7 9 8	16			
	Brown, moist, Silty fine to coarse SAND (SM) with Gravel, Medium Dense	2	7 6 5 6	11	27.9		
5	Grayish brown, moist to wet, Silty fine to medium SAND (SM), Medium Dense	3	7 8 8 7	16			
	Dark brown, wet, Silty fine to medium SAND (SM), Medium Dense	4	2 3 3 4	6			
10		5	1 2 4 3	6			
	Grayish brown, wet, Silty fine to medium SAND (SM), Loose	6	1 2 4 3	6			
15		7	2 2 3	5	32.7		
20	Gray, wet, Silty fine to medium SAND (SM), Very Loose	8	WOH WOH WOH	WOH			
25		9	WOH WOH WOH	WOH			
30		10	WOH WOH WOH	WOH			
35		11	WOH WOH WOH	WOH			
40	Boring terminated at 40.0 ft.	12	WOH WOH WOH	WOH			



BORING LOG No. B-3

PROJECT: MRSB DST Storage Project

CLIENT: CFE IDIQ

PROJECT LOCATION: MCB Camp Lejeune, NC

LOCATION: See Attached Boring Location Exhibit

DRILLER: Catlin Engineers and Scientists

DRILLING METHOD: Mud Rotary

DEPTH TO - WATER> INITIAL: 6

AFTER 24 HOURS: 7

PROJECT NO.: G2023-186

NORTHING:

EASTING:

ELEVATION:

LOGGED BY: MCR

DATE: 11-20-23

CAVING> C

Depth (feet)	Description	Graphic Sample No.	Blow Counts	N-Value	% < #200	TEST RESULTS	
						Plastic Limit	Liquid Limit
	2 inches of Topsoil	1	3 5 6 7	11	9.1	10	20
	Brown, Poorly Graded fine to medium SAND (SP-SM) with Sand and trace Gravel, Medium Dense	2	6 5 6 5	11			
5	Brown and gray, moist, Silty fine to medium SAND (SM), Loose	3	2 2 3 2	5			
	Light brown, wet, Silty fine to medium SAND (SM), Loose	4	2 1 1 1	2	21.7		
10	Light grayish brown, wet, Silty fine to medium SAND (SM), Very Loose	5	4 5 6 5	11			
		6	2 4 5 7	9			
15		7	3 2 1	3			
20		8	WOH WOH WOH	WOH			
25		9	WOH WOH WOH	WOH			
30		10	WOH WOH WOH	WOH	46.6		
35		11	WOH WOH WOH	WOH			
40		12	WOH WOH WOH	WOH			



BORING LOG No. B-3

PROJECT: MRSB DST Storage Project

CLIENT: CFE IDIQ

PROJECT LOCATION: MCB Camp Lejeune, NC

LOCATION: See Attached Boring Location Exhibit

DRILLER: Catlin Engineers and Scientists

DRILLING METHOD: Mud Rotary

DEPTH TO - WATER> INITIAL: 6

AFTER 24 HOURS: 7

PROJECT NO.: G2023-186

NORTHING:

EASTING:

ELEVATION:

LOGGED BY: MCR

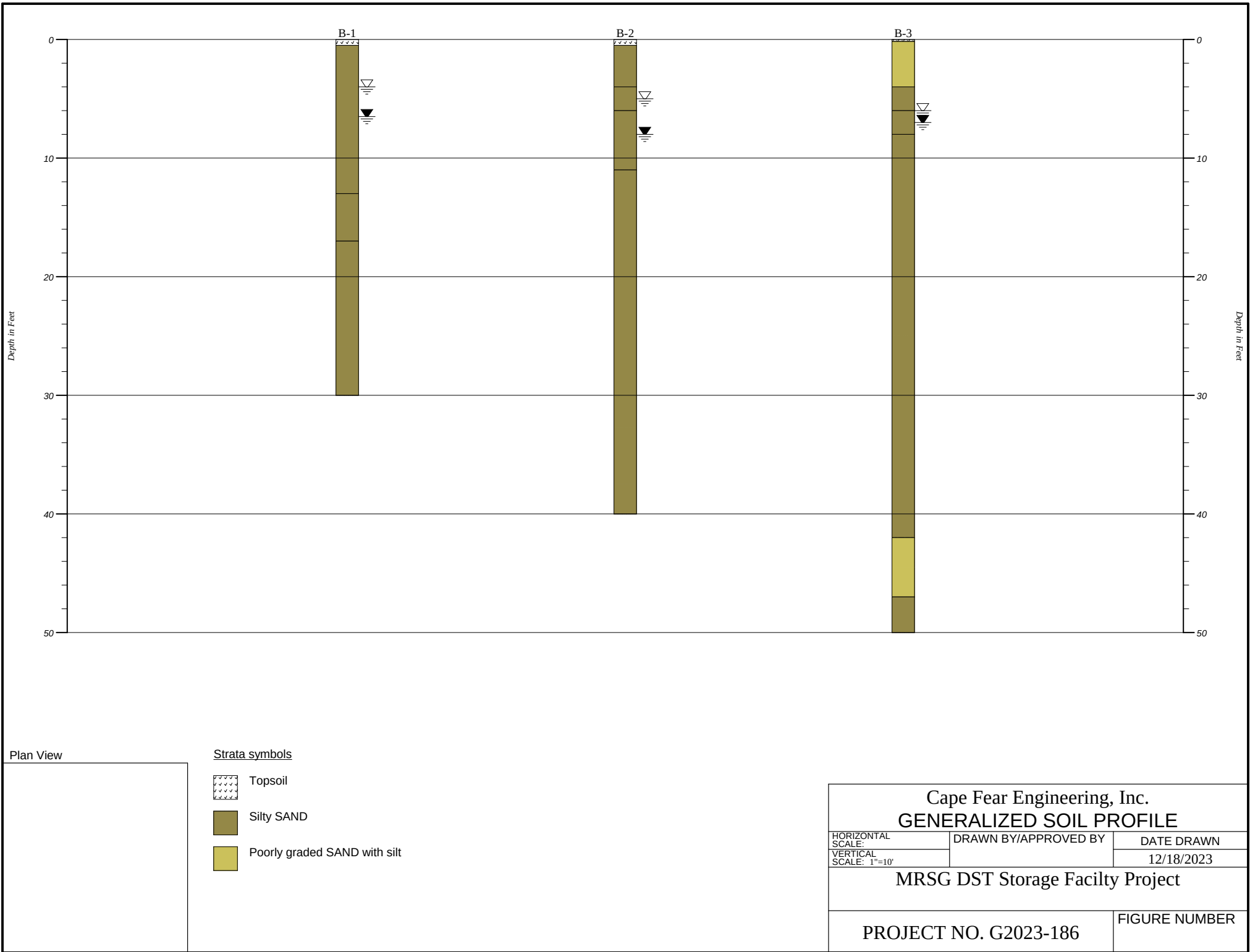
DATE: 11-20-23

CAVING> C

Depth (feet)	Description	Graphic Sample No.	Blow Counts	N-Value	% < #200	TEST RESULTS	
						Plastic Limit	Liquid Limit
						Water Content -	
						Penetration -	
						10 20 30 40 50	
	Light grayish brown, wet, Silty fine to medium SAND (SM), Very Loose (Continued)	42					
45	Dark green and light gray, wet, Poorly Graded fine to coarse SAND (SP-SM) With Gravel	13	26 50/1	100			100 →
	Brownish gray, wet, Silty fine to medium SAND (SM), Loose	47					
50		14	2 2 3	5			
	Boring terminated at 50.0 ft.						
55							
60							
65							
70							
75							
80							

APPENDIX IV

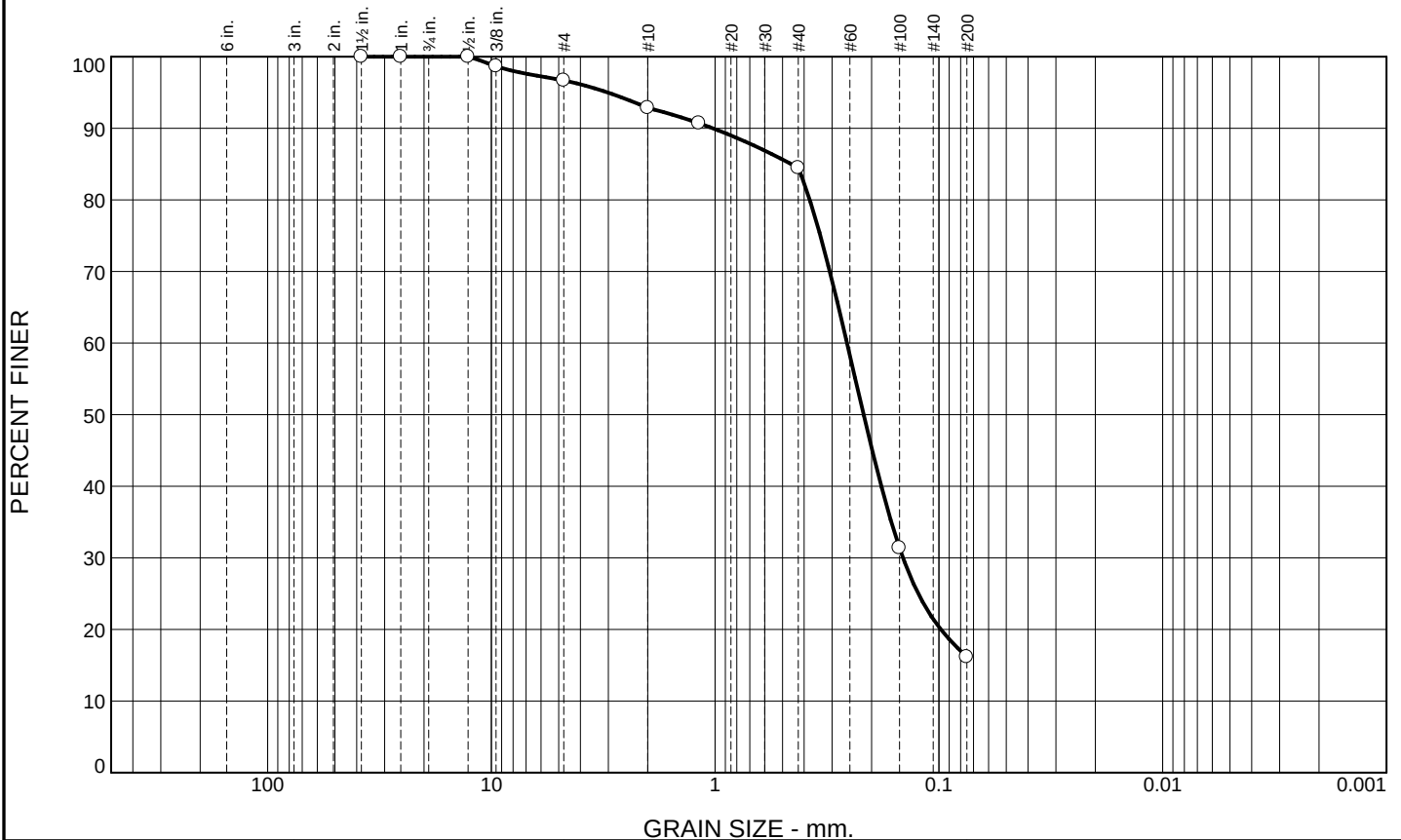
SOIL PROFILE



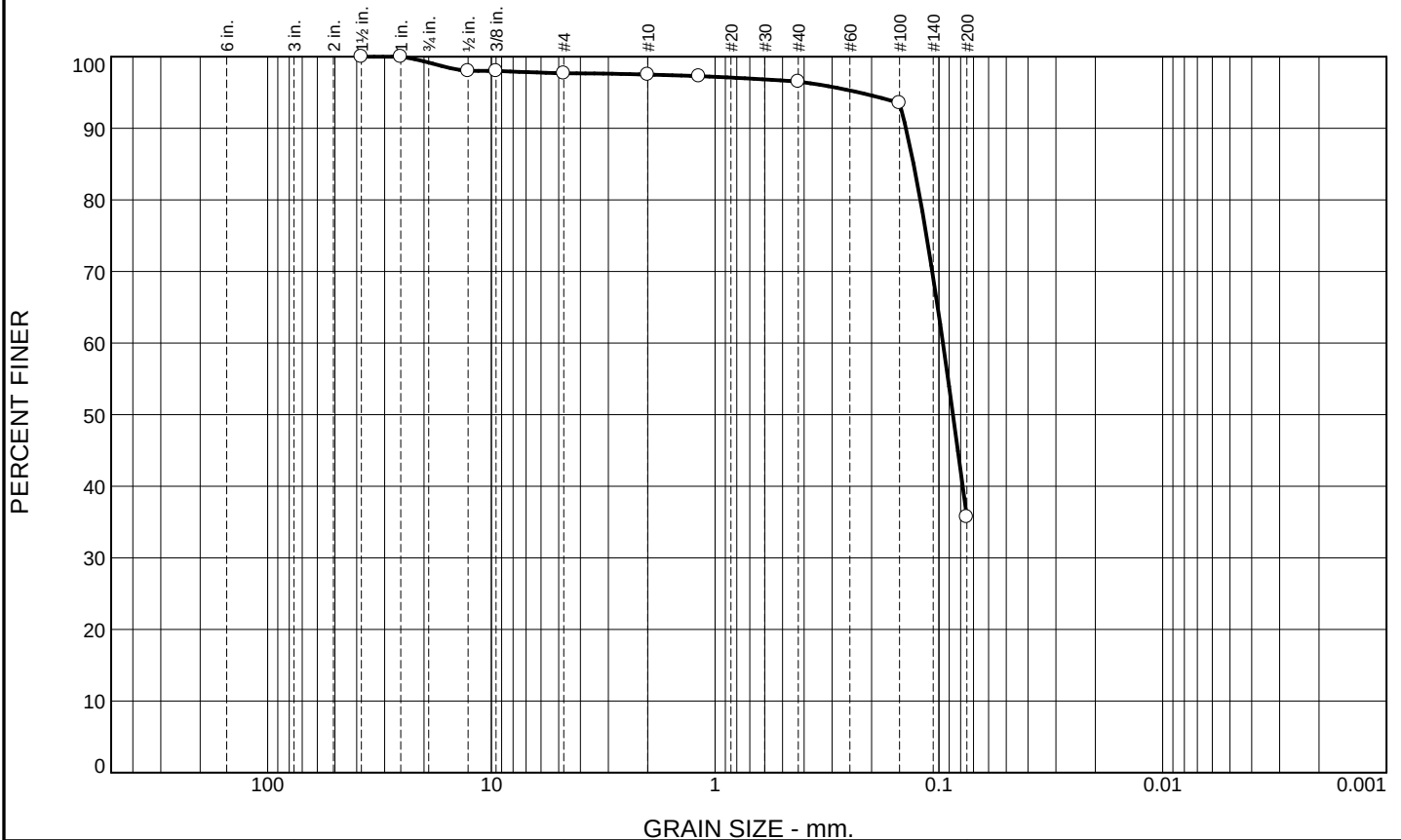
APPENDIX V

LABORATORY TEST RESULTS

Particle Size Distribution Report



Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.8	1.5	0.2	1.0	60.8	35.7	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	98.0		
.375	98.0		
#4	97.7		
#10	97.5		
#16	97.3		
#40	96.5		
#100	93.5		
#200	35.7		

* (no specification provided)

<u>Material Description</u>		
Grayish brown, Silty fine to medium SAND (SM) with trace Gravel		
<u>Atterberg Limits</u>		
PL= NP	LL= NV	PI= NP
<u>Coefficients</u>		
D ₉₀ = 0.1404	D ₈₅ = 0.1297	D ₆₀ = 0.0959
D ₅₀ = 0.0865	D ₃₀ =	D ₁₅ =
D ₁₀ =	C _u =	C _c =
<u>Classification</u>		
USCS= SM	AASHTO= A-4(0)	
<u>Remarks</u>		

Source of Sample: B-1 Depth: 18.5
Sample Number: 8

Date:



Client: CFE IDIQ
Project: MRSB DST Storage Project

Project No: G2023-186

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.6	7.0	4.7	9.5	50.3	27.9	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	97.8		
.375	96.4		
#4	92.4		
#10	87.7		
#16	84.5		
#40	78.2		
#100	50.6		
#200	27.9		

* (no specification provided)

Material Description

Brown, Silty fine to coarse SAND (SM) with Gravel

PL= NP **Atterberg Limits** LL= NV PI= NP

Coefficients
D₉₀= 3.0136 D₈₅= 1.2917 D₆₀= 0.2017
D₅₀= 0.1473 D₃₀= 0.0799 D₁₅=
D₁₀= C_u= C_c=

Classification
USCS= SM AASHTO= A-2-4(0)

Remarks

Source of Sample: B-2
Sample Number: 2

Depth: 2

Date:

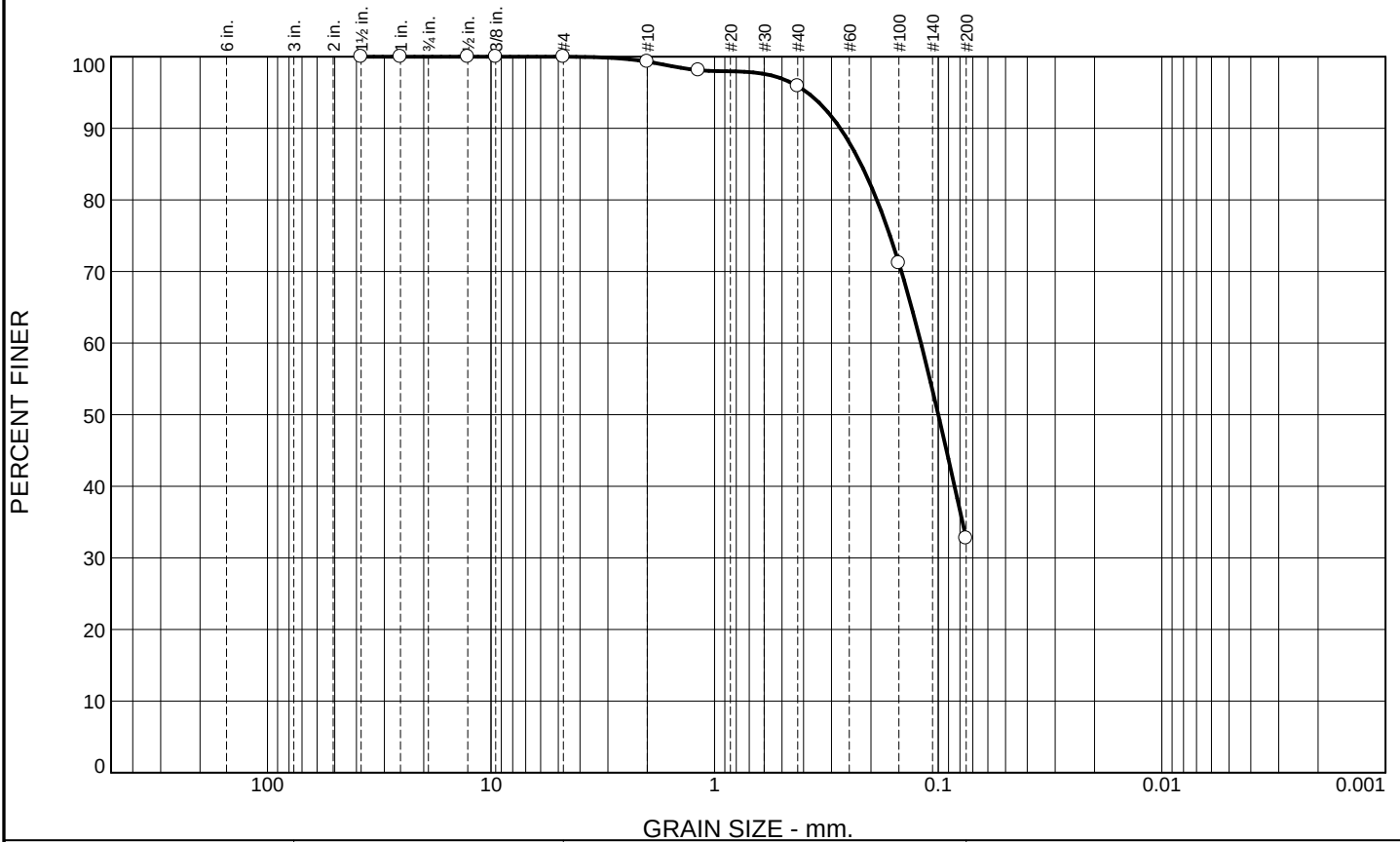


Client: CFE IDIQ
Project: MRSG DST Storage Project

Project No: G2023-186

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	0.0	0.7	3.4	63.2	32.7	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	100.0		
.375	100.0		
#4	100.0		
#10	99.3		
#16	98.1		
#40	95.9		
#100	71.2		
#200	32.7		

* (no specification provided)

Material Description		
Grayish brown, Silty fine to medium SAND S(M)		
Atterberg Limits PL= NP LL= NV PI= NP		
Coefficients D₉₀= 0.2742 D₅₀= 0.1000 D₁₀= D₈₅= 0.2215 D₃₀= C_u= D₆₀= 0.1195 D₁₅= C_c=		
Classification USCS= SM AASHTO= A-2-4(0)		
Remarks		

Source of Sample: B-2
Sample Number: 7

Depth: 13.5

Date:

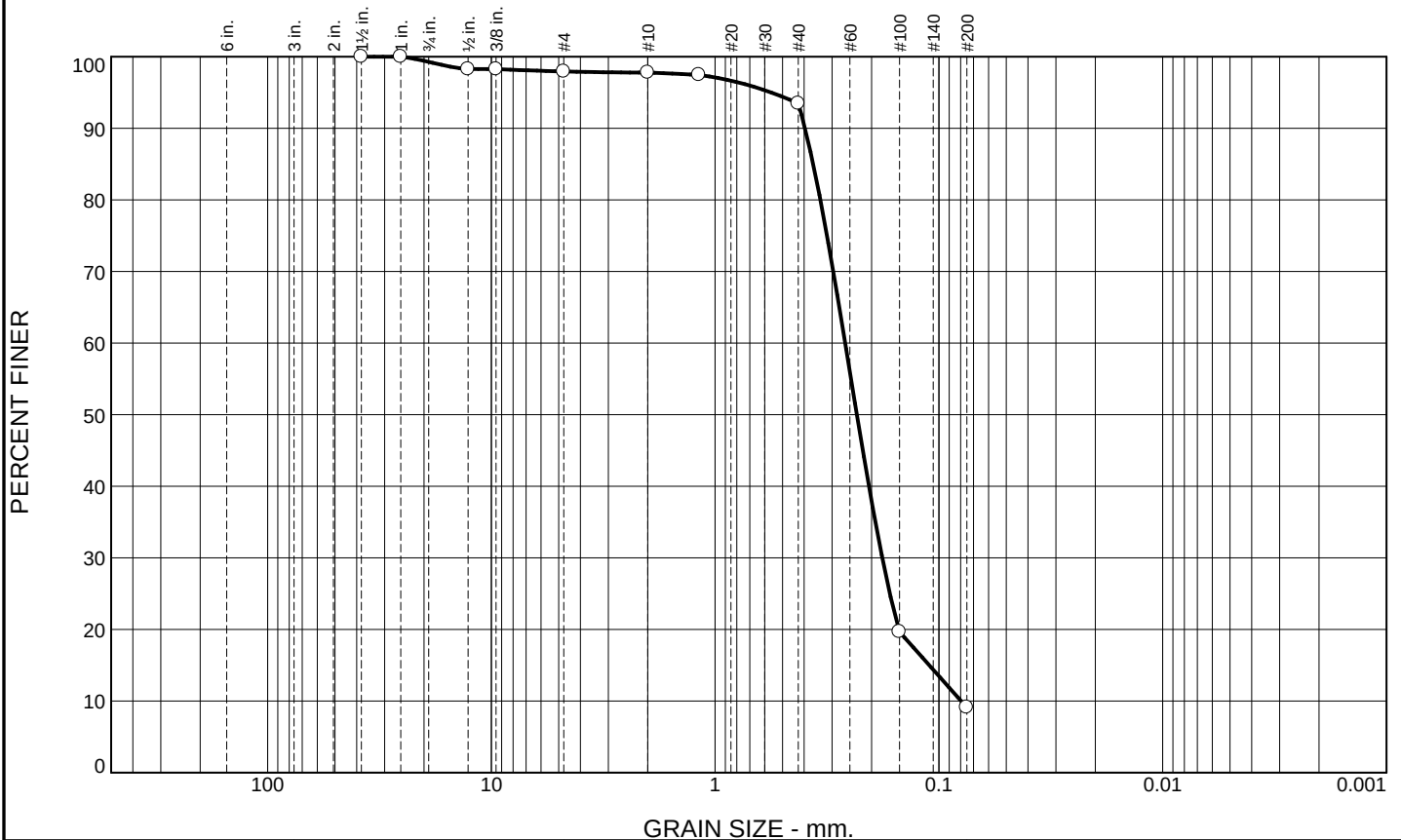


Client: CFE IDIQ
Project: MRSG DST Storage Project

Project No: G2023-186

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.7	1.4	0.1	4.3	84.4	9.1	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	98.3		
.375	98.3		
#4	97.9		
#10	97.8		
#16	97.5		
#40	93.5		
#100	19.7		
#200	9.1		

Material Description

Brown, Poorly Graded fine to medium SAND (SP-SM) with Sand and trace Gravel

Atterberg Limits

PL= NP

LL= NV

PI= NP

Coefficients

D₉₀= 0.3965

D₈₅= 0.3646

D₆₀= 0.2623

D₅₀= 0.2325

D₃₀= 0.1791

D₁₅= 0.1105

D₁₀= 0.0796

C_u= 3.30

C_c= 1.54

Classification

USCS= SP-SM

AASHTO= A-3

Remarks

* (no specification provided)

Source of Sample: B-3
Sample Number: 1

Depth: 0

Date:



Client: CFE IDIQ
Project: MRSG DST Storage Project

Project No: G2023-186

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	0.0	0.1	2.1	76.1	21.7	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	100.0		
.375	100.0		
#4	100.0		
#10	99.9		
#16	99.8		
#40	97.8		
#100	41.8		
#200	21.7		

* (no specification provided)

Material Description

Light brown, Silty fine to medium SAND (SM)

Atterberg Limits

PL= NP

LL= NV

PI= NP

Coefficients

D₉₀= 0.3497

D₈₅= 0.3170

D₆₀= 0.2087

D₅₀= 0.1760

D₃₀= 0.1077

D₁₅=

D₁₀=

C_u=

C_c=

Classification

USCS= SM

AASHTO= A-2-4(0)

Remarks

Source of Sample: B-3
Sample Number: 4

Depth: 6

Date:

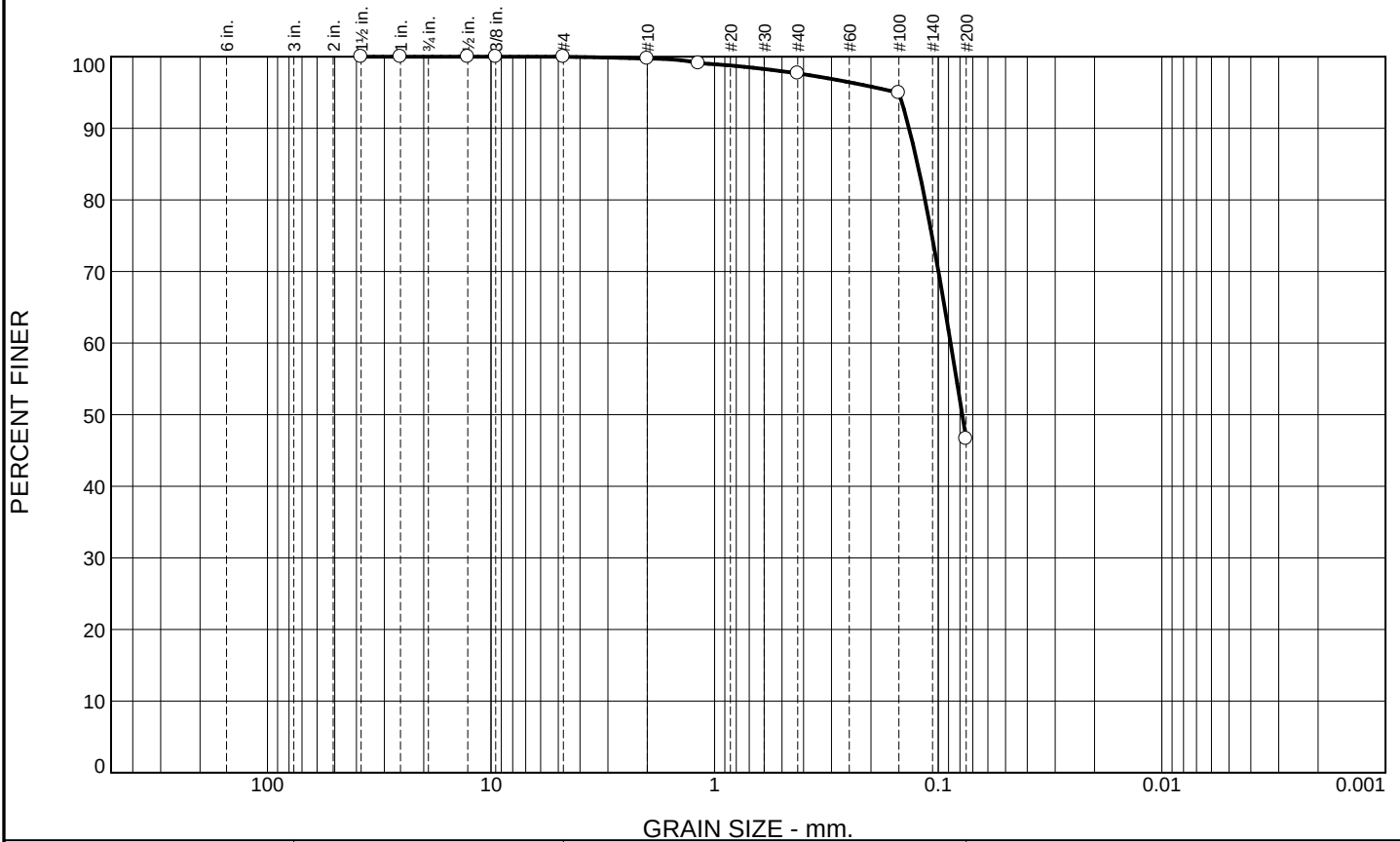


Client: CFE IDIQ
Project: MRSG DST Storage Project

Project No: G2023-186

Figure

Particle Size Distribution Report



% +3"	% Gravel		% Sand			% Fines	
	Coarse	Fine	Coarse	Medium	Fine	Silt	Clay
0.0	0.0	0.0	0.3	2.0	51.1	46.6	

SIEVE SIZE	PERCENT FINER	SPEC.* PERCENT	PASS? (X=NO)
1.5	100.0		
1	100.0		
.5	100.0		
.375	100.0		
#4	100.0		
#10	99.7		
#16	99.1		
#40	97.7		
#100	95.0		
#200	46.6		

* (no specification provided)

Material Description

Grayish brown, Silty fine to medium SAND (SM)

Atterberg Limits

PL= NP

LL= NV

PI= NP

Coefficients

D₉₀= 0.1350

D₅₀= 0.0780

D₁₀=

D₈₅= 0.1237

D₃₀=

C_u=

D₆₀= 0.0880

D₁₅=

C_c=

Classification

USCS= SM

AASHTO= A-4(0)

Remarks

Source of Sample: B-3
Sample Number: 10

Depth: 28.5

Date:



Client: CFE IDIQ
Project: MRSG DST Storage Project

Project No: G2023-186

Figure